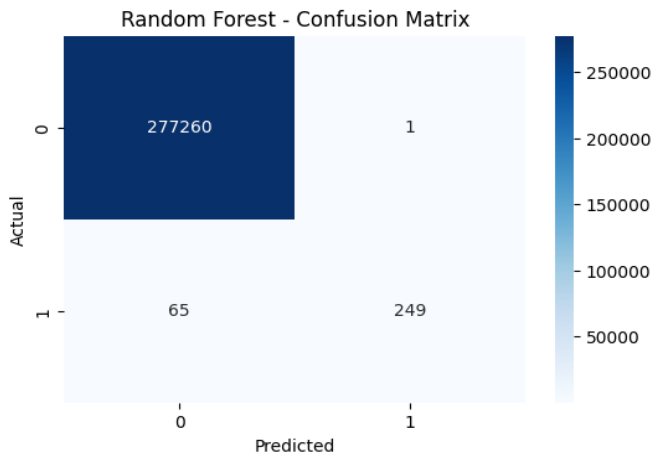
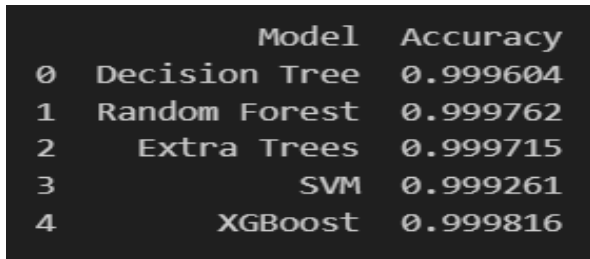


Project Development Phase Model Performance Test

Date	20 February 2026
Team ID	LTVIP2026TMIDS88068
Project Name	Online Payments Fraud Detection using Machine learning
Maximum Marks	4 Marks

Model Performance Testing:

S.No	Parameter	Screenshot												
1	Confusion Matrix	 Random Forest - Confusion Matrix <table><tr><th>Actual \ Predicted</th><th>0</th><th>1</th></tr><tr><th>0</th><td>277260</td><td>1</td></tr><tr><th>1</th><td>65</td><td>249</td></tr></table>	Actual \ Predicted	0	1	0	277260	1	1	65	249			
Actual \ Predicted	0	1												
0	277260	1												
1	65	249												
2	Accuracy Score	 <table><tr><th>Model</th><th>Accuracy</th></tr><tr><td>0 Decision Tree</td><td>0.999604</td></tr><tr><td>1 Random Forest</td><td>0.999762</td></tr><tr><td>2 Extra Trees</td><td>0.999715</td></tr><tr><td>3 SVM</td><td>0.999261</td></tr><tr><td>4 XGBoost</td><td>0.999816</td></tr></table>	Model	Accuracy	0 Decision Tree	0.999604	1 Random Forest	0.999762	2 Extra Trees	0.999715	3 SVM	0.999261	4 XGBoost	0.999816
Model	Accuracy													
0 Decision Tree	0.999604													
1 Random Forest	0.999762													
2 Extra Trees	0.999715													
3 SVM	0.999261													
4 XGBoost	0.999816													

3	Precision , Recall, F1 Score	<pre> Model Accuracy Precision Recall F1-Score 0 Decision Tree 0.999604 0.851724 0.786624 0.817881 1 Random Forest 0.999762 0.996000 0.792994 0.882979 2 Extra Trees 0.999715 0.991632 0.754777 0.857143 3 SVM 0.999261 1.000000 0.347134 0.515366 4 XGBoost 0.999816 0.985240 0.850318 0.912821 </pre>
4	Classification Report	<pre> Model Accuracy Precision Recall F1-Score 0 Decision Tree 0.999604 0.851724 0.786624 0.817881 1 Random Forest 0.999762 0.996000 0.792994 0.882979 2 Extra Trees 0.999715 0.991632 0.754777 0.857143 3 SVM 0.999261 1.000000 0.347134 0.515366 4 XGBoost 0.999816 0.985240 0.850318 0.912821 </pre>

Tune the Model

Parameter	Description
Hyperparameter Tuning	GridSearchCV used to optimize Random Forest parameters such as number of estimators and maximum depth
Validation Method	Train-Test Split (80% Training, 20% Testing) and Cross Validation